# Cybersecurity Incident Report

| **Section 1: Identify the type of attack that may have caused this network interruption** |
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| One potential explanation for the website's connection timeout error message is: a potential DoS attack might have took place since there has been a network flood on the website<br><br>The logs show that: there has been an attack activity<br><br>This event could be: a SYN DoS attack |

| **Section 2: Explain how the attack is causing the website to malfunction** |
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| When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol. Explain the three steps of the handshake:<br>1. The SYN packet is the initial request from an employee trying to connect to the web page hostess on the web server<br><br>2. The SYN ACK is the web server's response to the visitor's agreeing the connection<br><br>3. The ACK packet is the visitor's machine acknowledging the permission to connect<br><br>Explain what happens when a malicious actor sends a large number of SYN packets all at once: if the number of SYN requests is greater than the server's resources to handle the request, the server will be overwhelmed and unable to respond to the request.<br><br>Explain what the logs indicate and how that affects the server: two types of error, html/1.1 504 gateway time-out error and and RST, ACK packet |